SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0713

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

Code of Conduct:

I LOKESH S (192572151) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

LOKESH S

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.

2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

SCENARIO 1: SMART CAMPUS SUBNET ALLOCATION

Given Network

IP Address Block: 192.168.100.0/24

Subnet Allocation Table

Department	Subnet Mask	Network Address	Valid Host IP Range	Broadcast Address	Usable Hosts
School of Computing	/25 (255.255.255.128)	192.168.100.0	192.168.100.1 – 192.168.100.126	192.168.100.127	126
School of Electronics	/26 (255.255.255.192)	192.168.100.128	192.168.100.129 – 192.168.100.190	192.168.100.191	62
School of Mechanical Engineering	/27 (255.255.255.224)	192.168.100.192	192.168.100.193 – 192.168.100.222	192.168.100.223	30

1. **Network Address and Broadcast Address**

Department	Network Address	Broadcast Address
School of Computing	192.168.100.0	192.168.100.127
School of Electronics	192.168.100.128	192.168.100.191
School of Mechanical Engineering	192.168.100.192	192.168.100.223

2. **Valid Host IP Address Range**

- School of Computing: 192.168.100.1 – 192.168.100.126
- School of Electronics: 192.168.100.129 – 192.168.100.190
- School of Mechanical Engineering: 192.168.100.193 – 192.168.100.222

3. **Number of Usable Host Addresses**

- School of Computing: 126 Hosts
- School of Electronics: 62 Hosts
- School of Mechanical Engineering: 30 Hosts

4. **Remaining Unused IP Address Range**

Unused IP Range: 192.168.100.224 – 192.168.100.255
Total Unused Addresses: 32

SCENARIO 2: MULTINATIONAL SOFTWARE COMPANY

Given Network

IP Address Block: 10.10.0.0/16

Subnet Allocation Table

Department	Subnet Mask	Network Address	Valid Host IP Range	Broadcast Address	Usable Hosts
Software Development	/24	10.10.0.0	10.10.0.1 – 10.10.0.254	10.10.0.255	254
Quality Assurance	/25	10.10.1.0	10.10.1.1 – 10.10.1.126	10.10.1.127	126
Human Resources	/26	10.10.1.128	10.10.1.129 – 10.10.1.190	10.10.1.191	62
Executive Management	/27	10.10.1.192	10.10.1.193 – 10.10.1.222	10.10.1.223	30

1. **Network Address and Broadcast Address**

Department	Network Address	Broadcast Address
Software Development	10.10.0.0	10.10.0.255
Quality Assurance	10.10.1.0	10.10.1.127
Human Resources	10.10.1.128	10.10.1.191
Executive Management	10.10.1.192	10.10.1.223

2. **Valid Host IP Address Range**

- Software Development: 10.10.0.1 – 10.10.0.254
- Quality Assurance: 10.10.1.1 – 10.10.1.126
- Human Resources: 10.10.1.129 – 10.10.1.190
- Executive Management: 10.10.1.193 – 10.10.1.222

3. **Number of Usable Hosts**

- Software Development: 254 Hosts
- Quality Assurance: 126 Hosts
- Human Resources: 62 Hosts
- Executive Management: 30 Hosts

4. **Remaining Unused IP Address Range**

- Unused IP Range: 10.10.1.224 – 10.10.255.255

SCENARIO 3: IPv6 ADDRESSING

Given IPv6 Network 2001:0db8:abcd:0000:0000:0000:0000:0000/64

1. **Compressed IPv6 Address**

   2001:db8:abcd::/64

2. **Expanded IPv6 Address**

   2001:0db8:abcd:0000:0000:0000:0000:0000/64

3. **Network Prefix**

   - Network Prefix = 2001:db8:abcd:0000::/64
   - The first 64 bits represent the network portion, while the remaining 64 bits represent the host/interface identifier.

4. **Total Number of Host Addresses Supported**

   Host Bits = 128 − 64 = 64

   $$2^{64} = 18{,}446{,}744{,}073{,}709{,}551{,}616$$

   Total Hosts = 18,446,744,073,709,551,616

5. **Total Possible Host Addresses Available**

   18,446,744,073,709,551,616 (2^{64})

SCENARIO 4: ROUTING TABLE LOOKUP

Routing Table
Destination Network
192.168.1.0/24
192.168.2.0/24
192.168.3.0/24

Destination IP Address
192.168.2.45

1. **Destination Subnet**

   192.168.2.0/24

2. **Matching Routing Table Entry**

   192.168.2.0/24

3. **Next-Hop Network / Interface**

The router forwards the packet through the interface connected to the 192.168.2.0/24 network.

4. **Default Route**

   If no routing table entry matches the destination IP address, the router forwards the packet using: 0.0.0.0/0 (Default Route)

SCENARIO 5: COMPANY LAN CONFIGURATION

Administration LAN
Subnet: 192.168.10.0/26

- Network Address: 192.168.10.0
- Broadcast Address: 192.168.10.63
- Valid Host Range: 192.168.10.1 – 192.168.10.62
- Default Gateway: 192.168.10.1

Finance LAN
Subnet: 192.168.10.64/26

- Network Address: 192.168.10.64
- Broadcast Address: 192.168.10.127
- Valid Host Range: 192.168.10.65 – 192.168.10.126
- Default Gateway: 192.168.10.65

1. **Network Address and Broadcast Address**

LAN	Network Address	Broadcast Address
Administration LAN	192.168.10.0	192.168.10.63
Finance LAN	192.168.10.64	192.168.10.127

2. **Valid Host IP Address Range**

LAN	Host Range
Administration LAN	192.168.10.1 – 192.168.10.62
Finance LAN	192.168.10.65 – 192.168.10.126

3. **Suggested IP Addresses for Three Computers**

Administration LAN

- PC-1: 192.168.10.2
- PC-2: 192.168.10.3
- PC-3: 192.168.10.4

Finance LAN

- PC-1: 192.168.10.66
- PC-2: 192.168.10.67
- PC-3: 192.168.10.68

4. Default Gateway

LAN	Default Gateway
Administration LAN	192.168.10.1
Finance LAN	192.168.10.65